

Filtering, Personas, and the Limits of Synthetic Annotator Diversity

Can persona-prompted LLMs substitute for diverse human annotators?

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Abstract

Annotation disagreement is often treated as something to remove or something to simulate, but these two operations need not have the same statistical consequences. This study evaluates the published rater filter used for DICES-350 alongside a persona-prompting experiment designed to test whether demographic descriptions can reproduce the between-group variation found among human annotators. DICES-350 contains 350 conversations rated by 123 people in a fully crossed design, yielding 43,050 ratings, from which the dataset authors removed 19 raters after a post-hoc quality assessment.

Three results are robust. First, the filter’s effect on pooled label distributions is indistinguishable from removing 19 raters at random: mean total variation distance is 0.01556 and every permutation comparison is non-significant. With 123 raters behind each item, a 15.4% deletion is absorbed by ordinary sampling variation, which suggests replication depth as an important moderator of filtering impact rather than establishing that filtering is harmless. Second, metric choice determines the substantive conclusion about contested items: TVD rises monotonically across disagreement tertiles while JSD does not, because JSD weights each category by the inverse of its probability and amplifies near-empty cells. Third, persona-prompting did not produce demographic variation distinguishable from same-persona sampling variation. Using `qwen/qwen3.8-27b` under two prompt formulations, between-cell differences of 2/15 and 3/15 items fell inside a same-persona sampling floor measured on the same items (mean 2.10/15, range 1–4), and the model under-flagged unsafe content by roughly a factor of five relative to the human pool. Under a common one-draw-per-group estimator, human groups differ on 52% of items while both persona conditions remain within the noise floor.

Two further results concern gender and differ in evidential status. The filter removes men at 23.0% against 8.1% for women (odds ratio 3.36, 95% CI [1.05, 12.83], $p = 0.0262$): a planned comparison, since each demographic variable was specified for testing before any result existed, though it does not survive family-wise correction across the four demographic axes. Gender is also the only axis on which between-group spread falls relative to random removal ($p = 0.037$), but that comparison was selected after inspecting four groupings and is reported as suggestive only. Taken together the results do not support a general claim that filtering and persona-prompting flatten disagreement in the same way. Filtering leaves demographic structure largely intact at this replication depth, while the tested model generated no demographic variation to flatten.

1 Introduction

1.1 Disagreement as a question of whose judgment counts

A safety annotation is never only a statement about an item. When a rater marks a conversational response as safe, unsafe, or unsure, the label is also a statement about a judgment made by one person under a particular interpretation of the item. When disagreement is collapsed into a single majority label, that distribution of judgments becomes less visible. The central problem is therefore not simply whether an annotation is accurate, but whose judgments remain represented after aggregation. Aroyo and Welty (2015) frame disagreement as meaningful information about annotation rather than as a defect to be eliminated, while Basile et al. (2021) motivate evaluation procedures that preserve the structure of disagreement rather than treating a single gold label as the only valid target. The present study takes that premise seriously by examining two operations that act in opposite directions: removing raters considered low quality and asking a language model to simulate additional perspectives.

Rater filtering is usually framed as a cleaning operation. A subset of annotators is removed because their behavior appears inconsistent with the intended task. Quality checks of this kind typically target signals such as implausibly short completion times or near-constant labelling, though the criteria behind any particular published filter, including the one examined here, are often not stated. The intuitive benefit is that the remaining pool should provide more reliable labels. The corresponding risk is that the filter may remove a particular type of person disproportionately and thereby change the distribution of judgments. Fleisig et al. (2025) provide the core motivation for testing this risk: filtering can distort label distributions, particularly when labels are estimated from relatively small rater pools. Persona-prompting reverses the direction of the proposed intervention. Instead of deleting

people, it tries to create additional variation by asking a model to behave as a member of a specified demographic group. Pavlovic and Poesio (2024) develop this as direct representation, Hu and Collier (2024) measure how much persona conditioning actually moves model annotations, and Kirk et al. (2024) and Sorensen et al. (2024) connect it to personalized and pluralistic alignment.

The two practices therefore appear to be opposites: one removes variation and the other attempts to add it. Yet an observed difference between two persona conditions is not automatically evidence of a demographic effect. A model may produce different outputs because of ordinary sampling variability even when the persona text has no causal influence on the decision. Likewise, a filtered human pool may appear different simply because every finite subsample fluctuates. The common requirement is a baseline for sampling noise. A difference of two items out of fifteen cannot be interpreted as evidence of persona conditioning unless the researcher knows how often an unchanged persona produces a difference of two items. The same logic applies to between-group comparisons after filtering. This paper therefore treats the null model as part of the measurement itself rather than as a secondary significance test.

1.2 Research questions

The analysis is organized around three research questions.

RQ1.

Does the published quality filter applied to DICES-350 remove raters non-randomly with respect to demographics, and does that removal alter the resulting label distributions?

RQ2.

Does persona-prompting an LLM with demographic descriptions reproduce the group-level rating variation observed among human annotators?

RQ3.

Compared with the full human rater pool, do filtered-human labels and persona-prompted LLM labels exhibit reduced between-group spread, and if so, do they reduce it in the same way?

1.3 Contributions

- An audit of the filter actually applied to the DICES-350 rater pool rather than a threshold constructed specifically for this study.
- A scope condition showing why a demographically biased filter can be distributionally mild at high replication depth.
- A metric analysis showing that TVD and JSD can lead to different substantive conclusions about contested items.
- A persona-prompting evaluation with a measured same-persona sampling floor and an explicitly stated stop rule.
- A negative result on annotator substitution at both the aggregate-label and between-group-variation levels.

2 Background and Related Work

2.1 Disagreement as signal

The literature can be organized around different answers to a single question: what does disagreement mean? Aroyo and Welty (2015) treat disagreement as information carried by the crowd rather than as a nuisance that can simply be removed. In this view, disagreement can tell us something about

how a task is interpreted and about the diversity of judgments available to an aggregation procedure. Basile et al. (2021) extend this perspective to evaluation, motivating the use of distributions and disagreement-aware metrics rather than a single gold label as the only summary of human judgment. Kirk et al. (2024) and Sorensen et al. (2024) push the importance of disagreement into design and alignment, where the goal is not merely to estimate a majority position but to understand the range of values or judgments represented by the data.

These positions are related but not identical. If disagreement is viewed as noise, a quality filter appears desirable: the analyst seeks to remove unreliable variance. If disagreement is viewed as signal, removal is a substantive operation because it changes whose judgments enter the distribution. If disagreement is viewed as evidence about the item or about different interpretations of the task, then the correct response may be to preserve several structures at once rather than to treat all variation as one scalar quantity. The present work does not attempt to resolve these philosophical differences. Instead, it asks what happens empirically when a real published filter removes a fixed set of raters and when an LLM is asked to imitate demographic groups.

2.2 Taxonomies of disagreement

Asking what disagreement *is* produces more than one answer, and the answers do not reduce to a single decomposition. Four accounts are relevant here, and they differ in where they locate the disagreement rather than merely in vocabulary.

Lindahl (2024) distinguishes boundary, label, and existence disagreement: annotators may differ about where the task’s boundary falls, about which label applies within it, or about whether a valid answer exists at all. Jiang and de Marneffe (2022) work bottom-up from data rather than top-down from theory, developing a taxonomy of ten disagreement sources over MNLI and ChaosNLI items, grouped into three high-level classes. *Uncertainty in sentence meaning* covers lexical ambiguity, implicature, presupposition, probabilistic enrichment, and imperfection; *underspecification in guidelines* covers coreference, temporal reference, and interrogative hypotheses; *annotator behaviour* covers accommodating minimally added content and high lexical overlap heuristics. The three classes locate disagreement in the item, in the task definition, and in the annotator respectively.

Weber-Genzel et al. (2024) draw a different line, between human label variation (different labels assigned for valid reasons) and annotation error, where a label is assigned for an invalid reason. Their contribution is methodological as much as conceptual: in VariErr NLI they operationalise the distinction with a two-round procedure in which annotators first explain each label and then judge the validity of others’ label–explanation pairs, producing 7,732 validity judgments over 1,933 explanations for 500 re-annotated MNLI items. The distinction is therefore carried by the *reason*, not by the label. Sang and Stanton (2022) locate disagreement in the annotator instead, developing a multidimensional scale from interviews and concept mapping and testing it with 170 hate-speech annotators, and reporting that facets of individual difference such as age and personality relate to labelling decisions.

Table 1: Four accounts of disagreement, and what each implies about rater filtering.

Source	Domain	Categories	Located in
Lindahl (2024)	argumentation	boundary / label / existence	task, answer, or whether an answer exists
Jiang & de Marneffe (2022)	NLI	10 sources in 3 classes: uncertainty in sentence meaning; underspecification in guidelines; annotator behaviour	the item, the guidelines, or the annotator
Weber-Genzel et al. (2024)	NLI	human label variation vs annotation error, separated by validity of the stated reason	the reason behind the label
Sang & Stanton (2022)	hate speech	multidimensional scale of annotator individual differences	stable annotator characteristics

These accounts matter for the present study in a specific way, and the point is not one any of them

makes. A rater quality filter is an operation that removes annotators. Read against Jiang and de Marneffe’s taxonomy, such a filter is aimed at their third class, annotator behaviour, while their first two classes are precisely the signal a disagreement-aware analysis wants to keep. A filter that cannot distinguish the classes removes both. Read against Weber-Genzel et al., the difficulty sharpens: separating error from variation required them to collect explanations and validity judgments, because the distinction lives in the reason and not in the label pattern. The DICES filter, like most published quality filters, records no reasons at all. The error/variation distinction is therefore not merely unmeasured in this dataset but unavailable in principle, which is why Section 3.5 treats the published exclusion list as a single fixed operation rather than a criterion that could be re-derived.

Sang and Stanton’s finding adds a further caution. If disagreement is partly a function of stable annotator characteristics, then filtering on behavioural signals risks removing *people* rather than *errors*, which is testable and is what RQ1 tests.

This also fixes the scope of what this paper measures. The analysis operates on categorical item-level labels: soft-label vectors over No, Yes, and Unsure, compared with TVD and JSD. Under Lindahl’s decomposition this captures label disagreement directly but cannot separate boundary from existence disagreement, and under Weber-Genzel et al.’s it cannot separate variation from error at all. Two raters who disagree because they read the task differently are indistinguishable here from two who read it the same way and judged differently. The measurements therefore quantify one observable form of disagreement, and the negative persona result in particular is scoped to that form.

2.3 What to do with disagreement once it has been measured

Measuring disagreement and deciding what to do with it are separable problems, and the present study addresses only the first. Two lines of work address the second and are worth stating because they define what a measurement like this one is *for*.

Gordon et al. (2022) pose the question directly: whose labels should a model learn to emulate? Their answer, jury learning, replaces implicit majority vote with an explicit choice of which people or groups determine a prediction, and in what proportion. Architecturally they model every annotator in a dataset, sample from those models to populate a jury, then run inference. The framing is directly relevant here: a between-group spread measurement is a precondition for jury learning rather than a competitor to it. If a quality filter has silently removed one demographic axis from a corpus, a jury drawn from that corpus inherits the removal, and no amount of care in composing the jury will recover what is no longer there.

Romberg (2022) takes a different route to the same concern. Rather than changing whose labels count, PerspectifyMe enriches a prediction with a subjectivity score derived from the annotation process, so that a model reports both an aggregated hard label and an indication of which items elicited genuinely different views. Applied to argument concreteness, this treats contested items as something to flag rather than to resolve, an alternative to removing raters that leaves the disagreement visible downstream.

2.4 Rater quality and filtering

The project’s filtering analysis is motivated by the possibility that quality control can change the population that a dataset represents. Fleisig et al. (2025) are the central comparison because they connect filtering directly to distortion of label distributions, and because that distortion is most consequential when rater depth is low. The present study differs in a crucial structural way: DICES-350 has 123 raters per item. Consequently, deleting 19 raters removes 15.4% of the pool but still leaves 104 observations behind every item-level distribution. The question is therefore not whether a filter is demographically biased in the abstract; it is whether that bias is large enough, relative to replication depth, to produce a measurable change in the resulting distribution.

Homan et al. (2024) and Sang and Stanton (2022) provide additional motivation for examining intersectional and individual differences rather than treating demographic groups as interchangeable.

In the present analysis, the demographic audit is deliberately conducted on the published exclusion list. There is no quality flag in the dataset itself. Instead, the excluded identifiers appear in dataset documentation. This makes the analysis an audit of a real operation used in the released data rather than a simulation of an invented quality criterion.

2.5 LLMs as annotators and persona steering

The LLM arm is situated in a literature that considers whether language models can represent human characteristics directly or through persona conditioning. Pavlovic and Poesio (2024) propose direct representation, in which a model is asked to stand in for an annotator, while Hu and Collier (2024) quantify the effect of persona conditioning directly, reporting that persona variables account for under 10% of annotation variance in existing subjective datasets and that persona prompting yields modest though statistically significant gains. Kirk et al. (2024) and Sorensen et al. (2024) are connected to personalized and pluralistic alignment. The methodological gap highlighted by the present project is narrower: persona-conditioned differences need to be compared with a same-persona baseline. Without such a baseline, a between-group difference can be mistaken for evidence of demographic conditioning even when the model would have produced the same amount of disagreement under an unchanged persona.

The distinction is especially important for small evaluations. In the present pilot, fifteen items were used and each persona produced one draw per item. A change on two items may look substantial if the only reference point is zero change, but the same number is uninformative if a repeated draw from one unchanged persona commonly changes on two items. The project therefore measures the noise floor first, fixes the decision rule before the full run, and stops when the observed between-cell differences fail that rule.

2.6 Summary of the gap

Filtering and persona-prompting have each been examined, and disagreement has been taxonomised (Lindahl, 2024; Jiang & de Marneffe, 2022; Weber-Genzel et al., 2024), traced to annotator characteristics (Sang & Stanton, 2022), and modelled downstream (Gordon et al., 2022; Romberg, 2022). What has not been done is to measure both operations on the same items, with the same metric, against the same null, and to ask whether they converge on the same reduced view. That is the question this paper addresses.

3 Data

3.1 The DICES-350 dataset

DICES-350 contains 350 multi-turn adversarial conversations produced by human agents interacting with a generative dialogue model and annotated for safety by a demographically diverse rater pool. The dataset was released under CC BY 4.0. The primary annotation analyzed in this report is `Q_overall`, which has three possible responses: No, Yes, and Unsure.

3.2 Fully crossed annotation design

Every one of the 123 raters annotated every one of the 350 conversations. The resulting matrix contains 43,050 ratings, with no missing rater-item pairs and no duplicates. This fully crossed structure is important because each demographic group is evaluated over the same set of items. In a partially crossed corpus, one group can appear to disagree more simply because it was exposed to different items. Here, each group produces a complete profile over an identical item set, so between-group distributions can be compared directly without imputing unseen ratings.

The fully crossed design also clarifies what the published filter does. Because the same 19 raters are removed from every conversation, every item loses exactly 19 observations and the remaining item-level

Table 2: Core properties of DICES-350 used in the analysis.

Quantity	Value
Conversations	350
Raters	123
Ratings	43,050
Ratings per conversation	123
Q_overall: No / Yes / Unsure	0.611 / 0.327 / 0.063 (rounded)
safety_gold	175 Yes / 175 No
degree_of_harm	Extreme 199 / Moderate 98 / Benign 31 / Debatable 22

distribution is based on 104 raters. The filter therefore acts as a fixed thinning operation repeated across the entire matrix. This is precisely why treating the 350 items as independent realizations of the filter would be incorrect: the same exclusion decision generated all 350 item-level changes.

3.3 Rater pool

Table 3: Demographic composition of the full rater pool.

Gender	n	Race/ethnicity	n	Age group	n
Woman	62	White	30	gen z	56
Man	61	Black/African American	29	millennial	36
		Asian/Asian subcontinent	26	gen x+	31
		Latine/x	22		
		Multiracial	16		

Education was recorded as college or above for 75 raters, high school or below for 41, and other for 7. All 123 raters provided complete demographic information on the columns used in the main analysis. A 14-level raw race variable exists for robustness checks but is not used as the primary grouping.

3.4 Labels and item metadata

Across all 43,050 ratings, Q_overall is 61.1% No, 32.7% Yes, and 6.3% Unsure. Unsure is retained as a separate category throughout the main analysis. This decision preserves an explicit uncertainty response instead of forcing it into a binary classification. A robustness analysis collapses Unsure into No; the resulting mean TVD is 0.01187 compared with 0.01556 for the original three-way representation, the Spearman correlation between the two per-item series is +0.828, and the substantive conclusion is unchanged.

The safety_gold item field is exactly balanced: 175 Yes and 175 No. The degree_of_harm field contains 199 Extreme, 98 Moderate, 31 Benign, and 22 Debatable items. Extreme and Moderate are the primary higher-count categories in breakdowns; Benign and Debatable remain visible but are interpreted cautiously because of their smaller sample sizes. Harm_type contains 71 comma-joined values that were grouped into ten fine and four coarse categories using a first-tag rule fixed before inspecting results.

3.5 The published rater filter

The dataset authors removed 19 of the 123 raters after a post-hoc quality assessment and reported results on the remaining 104. The documentation states only that these raters failed data quality checks; it specifies no criteria, no per-rater reason, and no underlying score. Section 5.1.4 examines whether the removed raters resemble what such checks would plausibly target, but that characterisation is derived in this study and is not a published criterion. The exclusion decision is not stored as a

quality flag within the annotation table; instead, the 19 identifiers are listed in dataset documentation. The present study uses that published list directly. This choice prevents the analysis from becoming a threshold-search exercise in which the investigator selects a cut-off after inspecting the outcomes. The trade-off is that the individual reasons for exclusion cannot be reconstructed, so the published filter is treated as a single fixed operation.

4 Method

4.1 Soft labels and distance metrics

For every item and every rater set, the soft label is the empirical proportion of No, Yes, and Unsure responses. The primary distance measure is total variation distance (TVD), defined as

$$\text{TVD}(p, q) = \frac{1}{2} \sum_i |p_i - q_i|.$$

TVD has a direct interpretation as probability mass: a value of 0.01556 means that the two categorical distributions differ by 1.556 percentage points of mass on average. Jensen-Shannon divergence (JSD), computed with base-2 logarithms, is reported as a secondary metric.

Metric choice is not a cosmetic decision in this study because the two measures order contested items differently. For small perturbations, the JSD used here is approximated by

$$\text{JSD} \approx \frac{1}{8 \ln 2} \sum_i \frac{\Delta_i^2}{p_i}.$$

The approximation correlates at 0.931 with the observed JSD series, with a mean relative error of 5.3%. In this local approximation each squared change Δ_i^2 is weighted by $1/p_i$, so changes in low-probability categories receive disproportionately large influence. In this dataset, the consequence is concrete: item 140 ranks first by JSD but only 45th by TVD because an Unsure cell moves from two raters to zero. The mechanism was examined before the disagreement was resolved in favor of TVD; the choice of primary metric was therefore not a post-hoc attempt to obtain a preferred conclusion.

4.2 Between-group spread

For a chosen demographic grouping, each group produces a three-element soft-label vector for each item. Between-group spread is the mean pairwise TVD across groups, averaged over items. The analysis also records the mean distance of group distributions from the pooled distribution. These quantities answer slightly different questions. Raw spread describes how far group distributions are from one another, while excess spread compares the observed value with a size-matched permutation null. The latter is the key quantity for distinguishing demographic structure from finite-sample noise.

4.3 Null A: size-matched regrouping

Null A randomly reassigns raters to demographic cells while preserving the original cell sizes. The purpose is to estimate how much between-group spread would appear even if group membership contained no relationship to labels. This correction is necessary because raw spread cannot be expected to equal zero when groups contain only 10 to 56 raters. A small group produces an empirical distribution that fluctuates more than an infinite population parameter would. The meaningful benchmark is therefore observed spread minus its size-matched null expectation.

4.4 Null B: random removal of 19 raters

Null B addresses a different question: whether the published 19-rater filter does more than arbitrary thinning by the same number of people. The analysis samples 5,000 random sets of 19 raters and

compares the published filter with that empirical reference distribution. Null A uses 5,000 label permutations, and every null in the paper is computed from a single run so that repeated quantities agree exactly. The same logic prevents a common inferential error in the item-level analysis. A Wilcoxon test over 350 item shifts yields $p = 7.57\text{e-}07$, but that test is pseudo-replicated because all 350 observations depend on the same fixed removal set. The defensible unit for the filter operation is the rater-level removal set, not the item.

4.5 Persona-prompting protocol

The persona arm uses six demographic cells formed by three race categories (White, Black, Asian) crossed with two gender categories (Man, Woman). The grouping was not selected by visual inspection. Among the candidate grouping schemes, $\text{race3} \times \text{gender2}$ showed the largest excess over its size-matched null, with observed spread 0.20315, null 0.16549, excess +0.03766, and $z = +3.78$. Age3, by contrast, fell below its null ($z = -0.34$), so an age-based persona experiment would have targeted a demographic axis for which the human data showed no detectable signal.

Three distinct personas were defined within each selected cell. They differed on age and education, the demographic attributes not fixed by the race-by-gender cell, and were grounded in the most common real profiles present in the corresponding human cell. A no-persona baseline was also used. Every persona and baseline prompt explicitly framed the model as one specific individual rather than as an average member of a group. This instruction is conservative with respect to the hypothesis: it encourages persona-specific variation rather than suppressing it. Each record stores the verbatim persona paragraph together with SHA-256 hashes of the system and user texts, allowing the analysis to distinguish failure of the model to condition on a persona from failure to transmit the persona to the API.

4.6 Model and decision rule

The model requested was `qwen/qwen3.8-27b`, served through Groq. The API reports the same string as `model_version` and provides no dated snapshot; a silent model change would therefore not be pinnable from the recorded version string alone. Model selection was made from free-tier quota arithmetic rather than capability assessment. The experimental configuration used temperature 1.0, a maximum of 8 output tokens, and a one-word forced answer format.

Before the persona data collection, the study fixed a stop rule: the between-cell difference had to clear the same-persona noise floor at $P(\text{noise} \geq \text{observed}) \leq 0.05$ and also show the expected direction relative to the human contrast. The first implementation compared the observed difference with the mean of the noise floor, which was recognized as too weak because a random draw can exceed a mean about half the time. The criterion was corrected to a tail-probability rule before conclusions were made. The corrected rule reversed the interim decision from proceeding to stopping.

5 Results

5.1 RQ1: who gets removed, and what moves

5.1.1 Demographic pattern in removals

Gender is the only demographic axis with a statistically detectable imbalance. The Freeman-Halton exact test gives $p = 0.0262$ for gender, with Cramér’s $V = 0.206$. Men have an estimated removal odds ratio of 3.36 relative to women, with a 95% confidence interval of [1.05, 12.83]. The corresponding exact omnibus p -values for race/ethnicity, age group, and education are 0.1667, 0.4523, and 0.5907. The zero-removal Latine/x cell (0/22) has a nominal uncorrected p -value of 0.0232 in its pairwise comparison, but it does not survive correction for five tests and sits under a null omnibus result; it is therefore treated descriptively only.

Table 4: Removal counts and rates by demographic group.

Variable	Group	Pool	Removed	Rate
Gender	Woman	62	5	8.1%
	Man	61	14	23.0%
Age	gen z	56	7	12.5%
	millennial	36	8	22.2%
	gen x+	31	4	12.9%
Race	White	30	5	16.7%
	Black	29	6	20.7%
	Asian	26	5	19.2%
	Latine/x	22	0	0.0%
	Multiracial	16	3	18.8%

The size of the uncertainty matters, and two qualifications apply. First, nineteen removals are spread across a rater pool of 123, and the smallest race cell contains only 16 people. The confidence intervals are wide, with half-widths ranging from 7.6 to 20.8 percentage points (median 13.1) across the table, so the exact magnitude is not estimated precisely; the odds-ratio interval [1.05, 12.83] only barely excludes 1.

Second, this is one of four omnibus tests, and the correction is worth stating carefully because the four tests here are *planned*. The analysis protocol specified in advance that removal would be tested against each demographic variable, before any result existed; gender was not selected after inspecting which axis came out lowest. Applied as a single family, both Bonferroni and Holm raise $p = 0.0262$ to an adjusted 0.105, so the result does not survive family-wise correction. Whether a family-wise correction is the right instrument here is contestable: the four tests address four distinct pre-specified hypotheses about different variables rather than repeated attempts at one hypothesis, and on that reading each is legitimately evaluated on its own. The honest summary is therefore that the removal imbalance is a planned comparison significant at its nominal level, not significant after family-wise correction, and not a result selected for being the largest of several.

This is a weaker qualification than the one attached to the between-group comparison in Section 5.3.2, which was chosen after seeing four groupings and therefore carries the full post-hoc discount. The two gender results should not be read at the same evidential standard. Importantly, neither implies that the gender difference is itself the intended criterion; the analysis observes correlation with the published filter rather than recovering an individual causal reason for removal.

5.1.2 The demographic bias does not reach the pooled labels

Removing the 19 published raters changes the item-level soft distributions, but the observed magnitude is indistinguishable from that produced by randomly deleting any 19 raters. Mean TVD is 0.01556, with median 0.01403 and maximum 0.04339. The mean change in $P(\text{Yes})$ is +0.00452, moving the pooled proportion from 0.32669 to 0.33121. Nine of 350 majority labels flip under the tie-aware rule, corresponding to 2.6% of items; eight of those nine move toward Yes.

Table 5: Published filter compared with random removal of 19 raters (Null B).

Statistic	Observed	Null mean	z	p
Mean JSD	0.000550	0.000575	-0.16	0.4495
Mean signed $\Delta P(\text{Yes})$	+0.00452	+0.000003	+0.59	0.5635
Majority flips	9	7.40	+0.71	0.4764

The correct scope condition is therefore specific rather than universal. At 123 raters per item, the deletion of 19 people, even when demographically non-random, is not large enough to create a pooled-

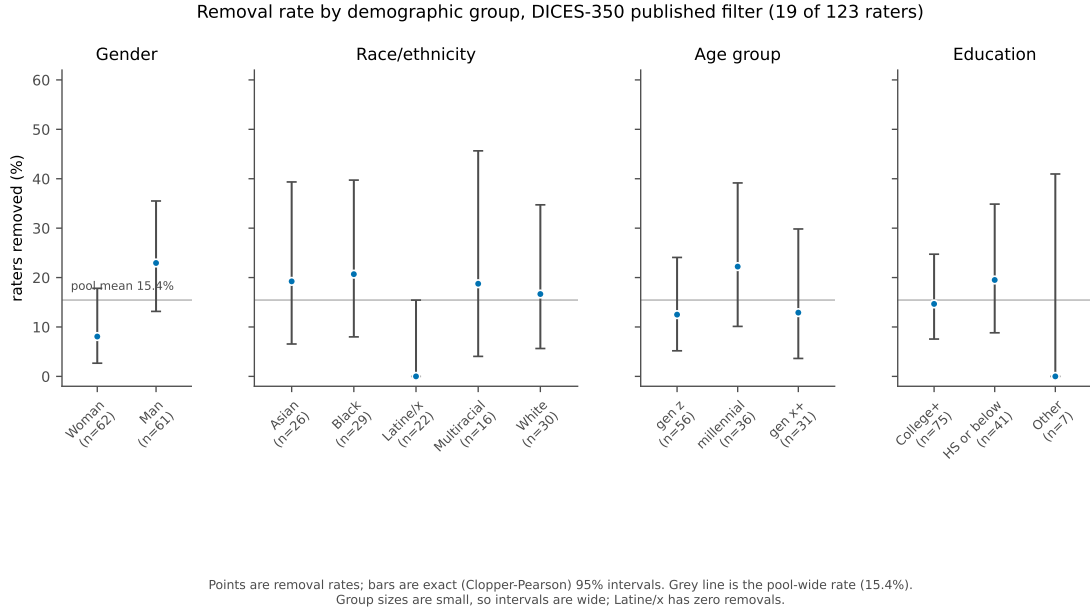


Figure 1: Removal rate by demographic group under the published DICES-350 filter (19 of 123 raters). Points are removal rates and bars are exact Clopper-Pearson 95% intervals; group size is given on each label. The grey line marks the pool-wide rate of 15.4%. Intervals are wide because group sizes are small, and no Latine/x rater was removed.

label shift that is distinguishable from ordinary finite-sample variation. The result should not be read as evidence that demographic filtering is harmless in general. It shows that the impact of this particular filter is bounded by the depth of the rater pool. With many observations behind every item, the soft-label estimator absorbs much of the perturbation.

5.1.3 Metric choice changes the contested-item conclusion

Table 6: Filtering effect across disagreement tertiles under the primary and secondary metrics.

Disagreement tertile	n	Mean JSD	Mean TVD	Mean $ \Delta P(\text{Yes}) $	Flips
Low	118	0.00077	0.01281	0.00922	0
Medium	115	0.00044	0.01580	0.01211	0
High	117	0.00044	0.01810	0.01431	9

TVD rises monotonically from 0.01281 in the low-disagreement tertile to 0.01580 in the medium tertile and 0.01810 in the high tertile, with a Kruskal-Wallis p -value of $8.4\text{e}-05$. The absolute change in $P(\text{Yes})$ follows the same ordering, as do the majority flips: 0, 0, and 9. JSD alone does not follow this pattern; its corresponding p -value is 0.49. The difference is not mysterious. The JSD approximation is weighted by the inverse of the category probability, so a small Unsure cell can dominate a change even when the total mass moved is modest. Item 140 illustrates this directly: it ranks first by JSD but 45th by TVD, because its Unsure count falls from two raters to zero.

This finding matters because metric selection can determine the substantive story a reader takes away. A JSD-only analysis would not support the claim that filtering has a stronger effect on contested items, while TVD, mean absolute change in $P(\text{Yes})$, and majority flips converge on that interpretation. The lesson is not that JSD is invalid, but that distributional conclusions depend on the geometry chosen to quantify change. In a disagreement-sensitive setting, the metric is part of the scientific argument and should be justified before results are interpreted.

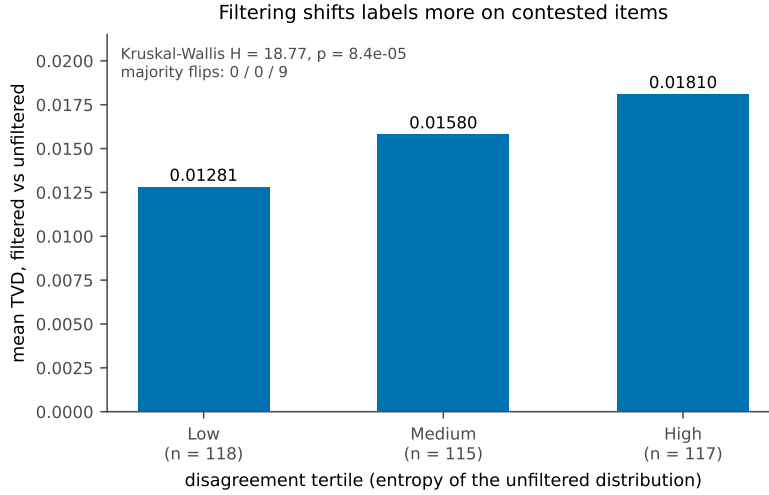


Figure 2: Mean TVD between filtered and unfiltered item distributions by disagreement tertile ($n = 118, 115, 117$). The increase is monotonic, Kruskal-Wallis $p = 8.4e-05$, and is consistent with the tie-aware majority-flip pattern of 0, 0 and 9.

5.1.4 Removed raters display quality signals

The rater-level profiles also support the interpretation that the published exclusion list is based on quality-related signals rather than on demographic selection. The removed group has a median answer time of 41,118 ms compared with 123,360 ms for the 104 kept raters, with a Mann-Whitney U p -value of 0.0013. Their modal-label share is higher (0.7143 vs 0.6343, $p = 0.0194$), while their label entropy is lower (0.8915 vs 1.0467 bits, $p = 0.0189$). Eighteen of 19 removed raters are extreme on at least one of these signals. Eleven of 19 have median answer times below the minimum observed in the kept pool, 45,162 ms, and one rater gave an identical answer across all 350 conversations and all 20 sub-questions.

These descriptive results corroborate the character of the published list but do not validate the underlying criteria. The threshold rules were never published, so the study cannot determine whether a specific individual was removed because of one signal, another signal, or a combination. What can be concluded is narrower: the published removals resemble the type of behavior the quality assessment was designed to detect, while the same list is also demographically skewed with respect to gender.

5.2 RQ2: persona-prompting

5.2.1 Selecting a human signal before testing the model

Table 7: Candidate grouping schemes scored against the size-matched null (Null A).

Grouping scheme	Cells	Min cell	Observed spread	Null	Excess	z
race3 $\times$ gender2	6	10	0.20315	0.16549	+0.03766	+3.78
gender2 $\times$ age3	6	14	0.16156	0.13989	+0.02167	+2.67
gender2	2	61	0.09170	0.07869	+0.01301	+1.26
race3	3	26	0.12887	0.11538	+0.01349	+1.21
race5	5	16	0.13835	0.12676	+0.01159	+1.40
age3	3	31	0.09603	0.09913	−0.00310	−0.34

The selection procedure produces two substantive observations before the model is queried. First, age does not show a human between-group signal in this dataset: its observed spread is slightly below the size-matched null, with $z = -0.34$. A persona experiment based primarily on age would therefore have risked constructing a test around a contrast the human data itself does not support.

Second, the intersectional race-by-gender grouping shows the strongest excess above its null, with $z = +3.78$. Because this grouping was selected from the observed candidate contrasts, the resulting human benchmark should be read as a data-selected contrast rather than as an independently specified demographic hypothesis; the comparison against the model is correspondingly optimistic about how much human structure was available to reproduce. Within the selected cells, Asian-Woman raters return 47.8% Yes responses over all items against White-Man raters’ 18.1%, a wider contrast than any corresponding single-axis comparison.

The chosen cells cover 85 of the 123 human raters. Latine/x (22) and Multiracial (16) are omitted from the six-cell primary grouping. This is important when interpreting the resulting benchmark: Multiracial is the group with the highest overall $P(\text{Yes})$, so the selected cells span a narrower range than the complete human pool. These two adjustments run in opposite directions: selecting the grouping with the largest observed excess inflates the benchmark, while excluding the two highest- $P(\text{Yes})$ groups deflates it. Their sizes are not known, so they cannot be netted against one another and neither is treated as cancelling the other. The resulting human contrast is therefore conservative with respect to the amount of demographic variation available in the source data.

5.2.2 Execution and provenance

The pilot returned 60 of 60 calls with zero API errors, zero refusals, and a 100% parse rate. In the full diagnostic stage, 180 calls were spent out of the planned 3,675-call budget before the stop rule was triggered. The mean input size was 922.9 tokens. The anticipated operational risk was refusal on adversarial safety content, but no such refusal occurred. Provenance checks passed, with each persona stored verbatim and the system and user prompt texts hashed.

5.2.3 Measuring the same-persona noise floor

To establish how much variation the model produces without any change of persona, one unchanged persona was sampled five times on the same 15 items. This yields an empirical same-persona sampling distribution, referred to below as the same-persona noise floor; it is a distribution of observed disagreement counts rather than a bound. Eleven of the fifteen items were identical across all five draws. A single draw differed from its item mode 8% of the time. Ten same-persona pair comparisons produced disagreement counts of [1, 1, 1, 1, 2, 2, 2, 3, 4, 4] out of 15, with a mean of 2.10 disagreements. This is the relevant baseline because the intended claim is not merely that different model calls can disagree, but that demographic persona conditioning should create more structured disagreement than the model produces by ordinary sampling.

5.2.4 The persona effect does not clear the pre-specified floor

Table 8: Persona-conditioned between-cell differences and the same-persona noise floor.

Condition	Between-cell difference	$P(\text{noise} \geq \text{obs})$	Percentile	Directional gap
Prompt v1	2/15	0.60	40th	+0.067
Prompt v2	3/15	0.30	70th	+0.000
Humans (matched items)	–	–	–	+0.324

Neither prompt formulation meets the decision rule. Under v1, the between-cell difference is two items out of fifteen and the probability that same-persona sampling noise is at least that large is 0.60. Under v2, the difference rises to three items, but the corresponding tail probability is still 0.30. The directional comparison is also unhelpful: the v2 personas have identical $P(\text{Yes})$ values because the three differing items cancel, and two of those three item-level differences run opposite to the human contrast. The measured within-cell disagreement of 4/15 is larger than the between-cell difference of 2/15 in the same pilot, reinforcing that ordinary sampling noise is already large enough to explain the observed separation.

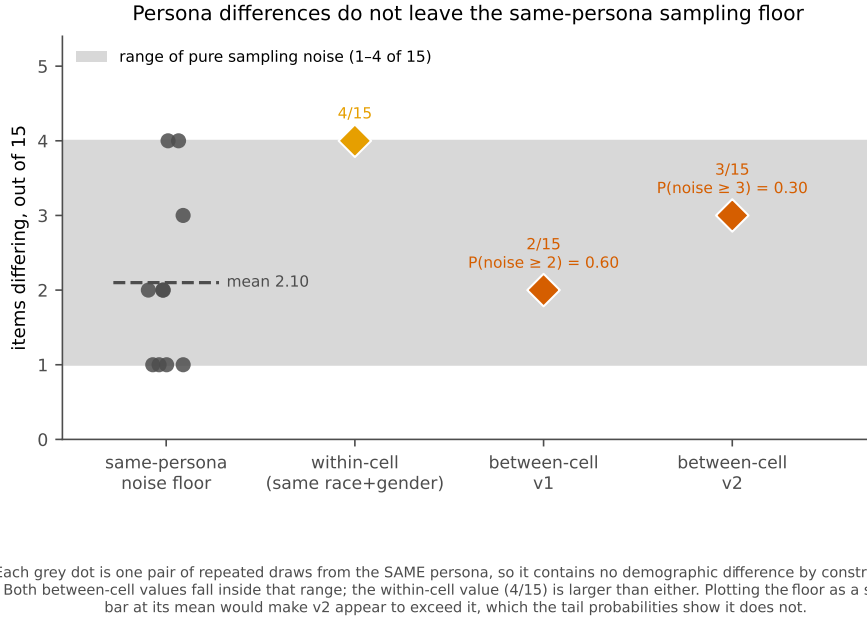


Figure 3: Persona differences against the same-persona sampling floor. Each grey point is one pair of repeated draws from the same persona and therefore contains no demographic difference by construction; the shaded band is their full range (1–4 of 15). Both between-cell values fall inside that band, with $P(\text{noise} \geq \text{observed})$ of 0.60 for v1 and 0.30 for v2. The within-cell comparison (4/15) is larger than either between-cell value.

The power limitation is explicit. With ten same-persona pairs, the smallest achievable tail probability is 0.10, so this pilot design cannot yield a p -value of 0.05 or less for a positive effect. It can, however, show that an observed between-cell difference is not distinguishable from the sampling variation observed under an unchanged persona, which is what occurs when the observed between-cell difference is small relative to the noise floor. That is what occurs here. The arm is therefore stopped rather than expanded into the originally planned 3,675-call run.

5.2.5 Robustness across prompt formulations

The result is not attributable to a single wording choice. Prompt v1 explicitly scoped the task to the model’s final turn. Prompt v2 replaced an earlier one-sided instruction with a two-sided formulation and moved the persona block adjacent to the rating instruction. The model’s $P(\text{Yes})$ changed from 0.050 to 0.044. Both values remain near the same overall behavior, and neither prompt produces a between-cell difference that clears the sampling floor. The materials therefore support the scoped claim that, under these two formulations, the tested model did not condition its DICES safety ratings on the demographic persona text.

5.2.6 The stronger negative: aggregate calibration also fails

Table 9: Unsafe-rate comparison on the matched 15-item subset.

Rater set	Matched 15	All 350
All 123 humans	0.278	0.327
Humans, Asian-Woman	0.433	0.478
Humans, White-Man	0.109	0.181
Model, v1	0.050	–
Model, v2	0.044	–

The persona failure is accompanied by an aggregate mismatch that does not depend on variation at all. On the matched 15 items, the human pool assigns Yes with probability 0.278. The Asian-Woman

human cell is at 0.433 and the White-Man cell at 0.109. The model assigns Yes only 0.050 under v1 and 0.044 under v2. Thus, on the matched items, the model is approximately 5.6 times less likely than the human pool to call an item unsafe, and it falls below even the most permissive human cell. This makes the negative result stronger than a simple failure to generate demographic variation: even before asking whether different personas disagree, the model’s overall calibration does not resemble the human annotation distribution used as its benchmark.

The scope of the claim remains narrow. The evidence supports a negative result for `qwen/qwen3.8-27b`, under two prompt formulations, on a 15-item matched evaluation. It does not establish that persona-prompting fails in general, nor does it establish that every language model would behave similarly.

5.3 RQ3: between-group spread

5.3.1 Negative control and the meaning of excess spread

Table 10: Between-group spread before and after filtering, with size-matched nulls.

Grouping	Condition	TVD	Null A	Excess	z
race3 $\times$ gender2	All raters ($n = 123$)	0.20315	0.16549	+0.03766	+3.78
race3 $\times$ gender2	Kept raters ($n = 104$)	0.20900	0.18352	+0.02548	+2.75
gender2	All raters ($n = 123$)	0.09170	0.07869	+0.01301	+1.26
gender2	Kept raters ($n = 104$)	0.08186	0.08508	−0.00322	−0.36
race5	All raters ($n = 123$)	0.13835	0.12676	+0.01159	+1.40
race5	Kept raters ($n = 104$)	0.15419	0.13669	+0.01750	+2.30
age3	All raters ($n = 123$)	0.09603	0.09913	−0.00310	−0.34
age3	Kept raters ($n = 104$)	0.10715	0.10727	−0.00012	−0.01

The age grouping serves as a negative control: its excess over Null A is close to zero in both conditions, at $z = -0.34$ across all 123 raters and $z = -0.01$ after filtering. In contrast the primary race3 $\times$ gender2 grouping reaches $z = +3.78$ before filtering and $+2.75$ after. The contrast shows that the measure separates demographic structure from finite-sample variation rather than reporting spread wherever groups are small. One qualification is necessary: age was identified as showing no excess over its null during the grouping-selection step on this same dataset, so it functions as a within-dataset consistency check rather than as an independently specified control. Raw spread by itself would be misleading: age has a raw TVD of 0.09603 even though its excess is essentially zero.

The filtering result also demonstrates why the null must be reported alongside the raw metric. Raw race3 $\times$ gender2 spread increases slightly from 0.20315 to 0.20900 after filtering, which might superficially look like an increase in diversity. Yet excess spread falls from +0.03766 to +0.02548 because the smaller post-filter cells are noisier. Filtering therefore makes the observed group distributions slightly farther apart in raw TVD while simultaneously reducing the amount of spread that can be attributed to demographic structure beyond sampling noise.

5.3.2 Filtering compresses one demographic axis

Table 11: Observed change in between-group spread relative to random 19-rater removal.

Grouping	Delta in spread	Null B	z	p
race3 $\times$ gender2	+0.00585	+0.01092	−0.42	0.642
gender2	−0.00984	+0.00606	−2.10	0.037
race5	+0.01584	+0.01031	+0.86	0.380
age3	+0.01112	+0.00895	+0.57	0.553

Gender is the only grouping for which the observed change is negative and more negative than the

random-removal reference. The gender excess moves from $+0.01301$ ($z = +1.26$) before filtering to -0.00322 ($z = -0.36$) after, and the mean group-to-pooled distance falls from 0.04585 to 0.04093.

Two qualifications are essential, and neither is minor. First, the pre-filter gender excess was itself not statistically significant ($z = +1.26$, $p \approx 0.21$). The correct description is therefore the disappearance of a weak trend, not the erasure of an established signal. Second, this observation is *not* independent of the RQ1 removal result. The same exclusion of 14 of 61 men against 5 of 62 women produces both: RQ1 measures who was removed, and this analysis measures what that removal did to the gender grouping. Their agreement is therefore close to one finding measured twice rather than two convergent tests.

What the comparison does add is directional. Removing raters shrinks every group, and smaller groups produce noisier distributions, which normally *increases* apparent spread, as the positive Null B means for the other three groupings show. The male group falls from 61 to 47 while the female group falls only from 62 to 57, so the naive expectation is more apparent gender spread, not less. Spread fell instead. The direction is therefore not a mechanical artefact of unequal thinning, even though the magnitude is small and the comparison is not independent.

The multiplicity qualification here is stronger than the one in Section 5.1.1, and the difference is not cosmetic. Four groupings were examined and gender was identified as the compressed axis *after* inspecting all four; the hypothesis that gender specifically would compress was not stated in advance. This is a post-hoc selected comparison, and it takes the full discount: a Bonferroni adjustment raises $p = 0.037$ to approximately 0.15, and it is reported as suggestive without further defence. The removal imbalance in Section 5.1.1 is a different case, a planned test of a pre-specified hypothesis, and the two should not be read at the same evidential standard.

5.3.3 Like-for-like comparison of human and model spread

The human group distributions and the LLM conditions initially have different estimator sizes. Human demographic cells contain 10 to 19 raters, whereas each LLM condition contributes one sampled response per group for each item. Directly comparing these quantities would favor the model because the one-sample estimator is highly variable and can produce extreme point-mass distributions. The analysis therefore reduces every condition to the LLM’s own estimator: one draw per group, with the human value estimated from 2,000 bootstrap draws.

Table 12: Between-group spread under a common one-draw-per-group estimator.

Condition	Scope	TVD	95% interval	$P(\text{noise} \geq \text{obs})$
Humans, 1 rater/group	15 items, 2 groups, $n = 1$	0.524	[0.267, 0.733]	0.00
LLM persona v1	15 items, 2 groups, $n = 1$	0.133	–	0.60
LLM persona v2	15 items, 2 groups, $n = 1$	0.200	–	0.30
Same-persona noise floor	15 items, $n = 1$	0.140	[0.067, 0.267]	–

With one categorical draw per group, each group distribution is a point mass, so the TVD between them is 1 when the two labels differ and 0 when they agree; the mean TVD is therefore exactly the proportion of items on which the two draws disagree. The comparison is stark but should be interpreted exactly as measured. Two randomly chosen humans from the selected demographic contrast disagree on 52% of the 15 matched items, yielding a TVD of 0.524. Two model-persona conditions yield 0.133 and 0.200. The same-persona noise floor is 0.140, with a range of [0.067, 0.267]. No same-persona pair reaches the human mean of 0.524, so $P(\text{noise} \geq \text{observed})$ is 0.00 for the human condition; that figure compares the ten same-persona values against the bootstrap mean. Taking the bootstrap distribution as a whole, 1,939 of 2,000 draws (96.9%) exceed the largest same-persona value, so the human contrast is not separated from sampling noise in every draw, only in the large majority of them. Both LLM conditions, by contrast, fall inside the range produced by unchanged-persona sampling. The model therefore does not demonstrate measurable demographic separation in the pilot in the sense required by the study design.

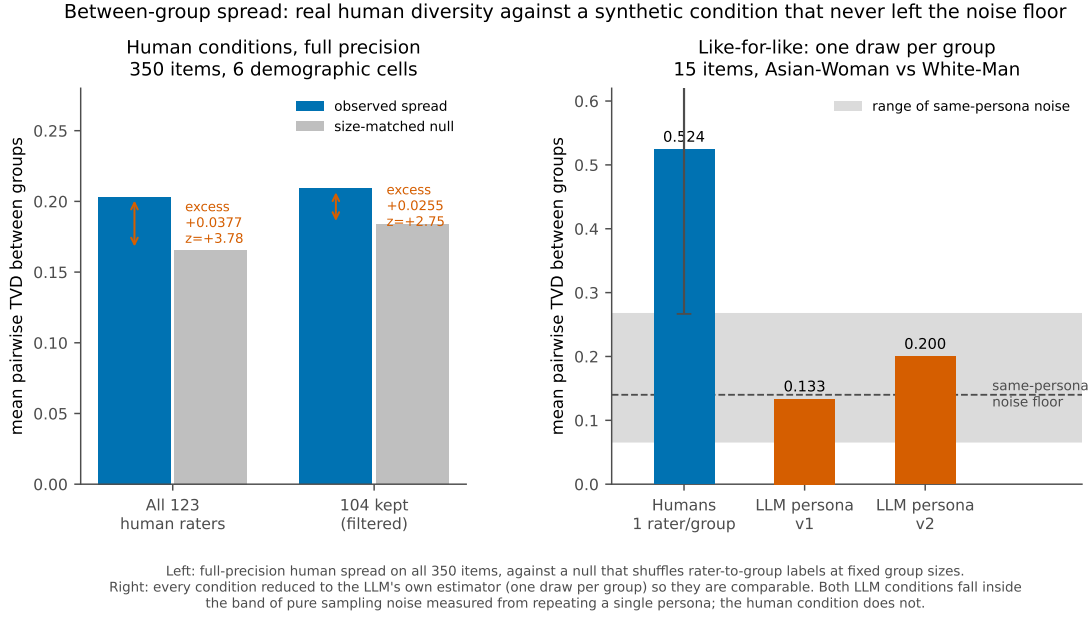


Figure 4: Between-group spread: human diversity versus the synthetic conditions. Left panel: full human spread over 350 items with a size-matched null. Right panel: like-for-like one-draw estimates on 15 matched items. The LLM persona conditions remain inside the same-persona noise range, while the human condition lies outside it.

This estimator-matched result also shows why the negative finding is not created by comparing a many-rater human distribution to a point estimate from the model. Once both sides use the model’s one-draw estimator, the human contrast becomes larger, not smaller. The evidence supports the conclusion that the tested persona prompts did not create group-level variation commensurate with the measured human contrast.

5.3.4 Breakdowns by disagreement and harm

Between-group spread rises across disagreement tertiles, from 0.17345 in the low group to 0.20302 in the medium group and 0.23323 in the high group. This is consistent with the interpretation that groups separate more on items for which raters disagree more generally.

By degree of harm the spread is comparatively flat, and it moves very little under filtering. Table 13 gives both conditions explicitly rather than as a range, since the direction of change matters and an unlabelled pair of numbers does not convey it.

Table 13: Between-group spread by degree of harm, primary grouping, before and after filtering.

Degree of harm	Items	All 123	104 kept	Change
Extreme	199	0.20305	0.20923	+0.00618
Moderate	98	0.20957	0.21665	+0.00708
Benign	31	0.18298	0.17875	−0.00423
Debatable	22	0.20388	0.21547	+0.01159

The Benign and Debatable categories are small (31 and 22 items) and are therefore reported without stronger interpretation.

6 Discussion

6.1 The hypothesis was partly right, and the correction is sharper

The initial framing suggested that rater filtering and persona-prompting could be viewed as opposite interventions that nevertheless narrow the range of represented judgments: filtering removes raters while persona-prompting attempts to recreate diversity synthetically. The results support only part of that framing, because the two interventions affect representation differently. Filtering changes the composition of the human rater pool but, at the observed replication depth, leaves the pooled distributions and most demographic structure largely intact. Persona-prompting, by contrast, did not reproduce the measured human between-group variation under the tested protocol. The distinction matters because it changes how each intervention should be diagnosed and where the risk lies.

Filtering does not flatten human diversity in general. After removing 19 raters, the primary $\text{race3} \times \text{gender2}$ grouping still has positive excess spread, $+0.02548$, with $z = +2.75$. The overall demographic structure therefore remains visible, and the pooled-label comparisons in Section 5.1.2 are indistinguishable from random removal. At this replication depth the answer to the original framing is that filtering does not produce a general narrowing of represented views.

Persona-prompting fails differently. Under the tested protocol there is no between-group variation distinguishable from same-persona sampling variation, so there is no demonstrated diversity for the comparison to flatten. Both persona conditions remain within the same-persona sampling floor, and the model simultaneously under-flags unsafe items relative to the human pool. This means the model cannot substitute for the human rater distribution under the tested protocol even before the more demanding question of whether it preserves demographic structure is considered. Unlike the filtering results below, this conclusion does not depend on any marginal significance test: the observed differences sit inside a directly measured noise floor.

A third, weaker observation concerns one specific axis. Gender is the only grouping whose between-group spread falls relative to random removal, moving from $z = +1.26$ to $z = -0.36$, with the groups also moving closer to the pooled position. This is the axis on which RQ1 finds the removal imbalance, so the two observations are consistent. They are not, however, equally strong, and they are not independent. The removal test was planned and is significant at its nominal level before family-wise correction; the compression comparison was selected after inspecting four groupings and carries the full post-hoc discount. Both are generated by the same exclusion set, so their agreement is closer to one finding measured twice than to two convergent tests. Targeted compression is therefore offered as a hypothesis the data are compatible with, not as an established result.

The sharper claim is thus that the two operations do not fail alike. Filtering changes who is represented while leaving most measurable demographic structure intact at this rater depth; the tested model did not produce evidence of demographic variation beyond the observed same-persona sampling variation. The strongest form of this paper’s conclusion does not depend on the gender result at all: at high replication depth a demographically skewed filter left pooled labels and overall demographic structure essentially intact, while a persona-prompted model produced no demographic variation distinguishable from its own sampling noise. Whether filtering additionally compresses one specific axis is a weaker, secondary question that this design can only answer suggestively.

6.2 Rater depth as a moderator of filtering impact

The most transferable result is the separation between demographic bias in a filter and measurable change in pooled labels. These are not the same outcome. DICES-350 has 123 raters per item, so a 15.4% deletion leaves a large number of independent observations behind each soft label. The resulting mean TVD of 0.01556 and the random-removal permutation comparisons show that the published filter’s pooled effect is absorbed by ordinary sampling variation at this replication depth. The filter can still be demographically biased without producing a detectable change in the overall item distribution.

This result is consistent with, rather than contradictory to, the filtering concern summarized from Fleisig et al. (2025). Their concern is relevant to settings with much shallower rater depth. Many annotation corpora collect only a handful of ratings per item, an order of magnitude below the replication depth available here. At such depths, the same deletion bias would act on distributions estimated from only a handful of observations, making the demographic composition of the remaining pool much more consequential. Because rater depth was not varied experimentally, this is a scope condition consistent with the present data rather than a demonstrated one: a biased filter can be statistically mild when replication is high and much more consequential when replication is low.

6.3 What a persona evaluation should measure before it scales

The persona arm suggests two methodological requirements. First, verify the human signal before asking a model to reproduce it. The age control demonstrates why this is necessary: age-based group differences are not stronger than the size-matched null in this dataset. A model that produced an age difference under a persona prompt would not automatically be reproducing a human pattern. Second, measure the same-persona noise floor. The observed 2/15 and 3/15 between-cell differences are not evidence of conditioning because unchanged personas produce an average of 2.10 disagreements out of 15. Without that baseline, the pilot could have been described as demonstrating persona diversity even though its variation was fully compatible with ordinary stochastic sampling.

6.4 Matching a statistic is not the same as representing a perspective

The final comparison also clarifies what a successful synthetic annotator would need to achieve. It would not be enough for a model to match one group’s average unsafe rate. A model could, in principle, reproduce a human cell’s aggregate $P(\text{Yes})$ while producing nearly identical item-level decisions across repeated persona draws. Such a model would imitate a mean without reproducing a perspective as a distribution of judgments. The human-versus-LLM spread metric in this project was designed to expose exactly this distinction. Kirk et al. (2024) and Sorensen et al. (2024) motivate pluralistic alignment, in which the range of human views matters rather than its central tendency. Under that framing, both the filtered pool and the synthetic substitute need to be evaluated for whether they preserve a meaningful distribution, not just an average.

7 Limitations

7.1 One dataset and one published filter

The analysis uses one dataset and one fixed published exclusion list. The filter contains 19 rater IDs, but the underlying threshold rules were never published. The study therefore cannot vary the filter, test alternative thresholds, or determine exactly which criterion was responsible for each individual removal. The conclusions concern this published operation rather than rater filtering as a general class of methods.

7.2 Replication depth limits the filtering conclusion

The null result for pooled labels is conditioned on 123 raters per item. The same demographic skew may have a larger effect in lower-depth datasets. The study consequently should not be generalized from “no detectable pooled effect here” to “demographic filtering is harmless.” The direction of the limitation is clear: this design is especially well suited to detecting a demographic imbalance in who is removed, but comparatively conservative for detecting how much that imbalance changes a pooled distribution when the remaining rater count is large.

7.3 Statistical power and multiplicity

Nineteen removals generate wide confidence intervals, with half-widths of 7.6 to 20.8 percentage points in Table 4. Both gender findings are affected by multiplicity, but they are *not* affected equally and are not reported at the same standard. The removal imbalance (Section 5.1.1, $p = 0.0262$) is a *planned* comparison: the protocol specified testing each demographic variable before any result existed. It does not survive family-wise correction across the four axes (Bonferroni and Holm both give an adjusted 0.105), but it was not selected for being the largest of several, and whether a single family-wise correction is appropriate across four distinct pre-specified hypotheses is contestable. The compression comparison (Section 5.3.2, $p = 0.037$) is *post-hoc*: gender was identified as the compressed axis after inspecting four groupings, so it takes the full discount and is reported as suggestive without further defence. The two are also not independent of each other, since the same exclusion set generates both. Likewise, the ten-pair same-persona pilot has a minimum achievable tail probability of 0.10. The pilot is informative for rejecting effects that clearly fail the noise-floor criterion, but it is not sized to estimate small positive persona effects precisely.

The same constraint appears at cell level. The smallest cells in the main analyses contain only 10 raters, which is enough to motivate a size-matched null but not enough to support precise demographic estimates. Every group-level rate is therefore reported with its N, and isolated small-cell differences are not interpreted as established effects.

7.4 Narrow cell coverage

The six race-by-gender cells used for persona selection cover 85 of 123 raters and exclude Latine/x and Multiracial participants. Because the Multiracial group has the highest overall $P(\text{Yes})$, the selected cells likely understate the total range available in the full rater pool. This makes the human benchmark conservative in the direction of the model: the model was not asked to reproduce the widest possible demographic contrast.

7.5 One model, two prompts, and no pinnable snapshot

RQ2 uses a single open-weights model selected by quota arithmetic, two prompt formulations, and 15 matched items in the decisive comparison. The API does not provide a dated model snapshot. A silent upgrade could therefore be invisible in the recorded version identifier. These limitations make the result model-level and protocol-level, not a general statement about persona-prompting. The study also stopped before the full 3,675-call run, so it does not provide a large-scale estimate of a small positive effect.

7.6 Only categorical label disagreement is measured

The study cannot separate the boundary, label, and existence forms of disagreement identified by Lindahl (2024). Every metric used here (soft labels, TVD, JSD, and between-group spread) assumes that disagreement is expressed as a categorical item-level label. Some of the measured spread may therefore reflect raters disagreeing about the question itself rather than simply giving different answers to a shared interpretation. The negative persona result is accordingly a result about this narrow, measurable form of disagreement.

8 Conclusion

This study examined filtering and persona-prompting as two different ways of changing the representation of disagreement in safety annotation. With 123 raters per item, the pooled-label changes produced by the published DICES-350 filter are indistinguishable from random removal of 19 people, and the primary demographic grouping retains clear excess spread after filtering ($+0.02548$, $z = +2.75$). At this replication depth the filter does not produce a general flattening of the measured human diversity.

The result therefore suggests that rater depth is an important moderator of how strongly demographic filtering affects pooled label distributions, though the low-depth case was not observed here. Two secondary observations concern gender and differ in strength. The removal imbalance (23.0% of men against 8.1% of women, $p = 0.0262$) is a planned comparison that is significant at its nominal level but does not survive family-wise correction across the four demographic axes. The reduction in between-group spread on the same axis was selected after inspecting four groupings and is suggestive only. Because the same exclusion set generates both, they are not independent, so targeted compression is offered as a hypothesis compatible with the data rather than as an established finding.

The persona arm failed a different test. Under two prompt formulations, `qwen/qwen3.8-27b` produced between-cell differences that remained inside the same-persona sampling floor. In the like-for-like one-draw comparison, the human contrast was 0.524 TVD, while the two persona conditions were 0.133 and 0.200. The model also under-flagged unsafe items relative to the human pool. Within the limits of one model, two prompts, and a 15-item pilot, the evidence therefore does not support treating persona prompting as a substitute for diverse human annotation. A useful next step is to repeat the same design at lower rater depth, where the filtering effect should become easier to observe, and to test multiple models with pinned snapshots and larger same-persona baselines.

References

- Aroyo, L., & Welty, C. (2015). Truth Is a Lie: Crowd Truth and the Seven Myths of Human Annotation. *AI Magazine*, 36(1), 15–24. <https://doi.org/10.1609/aimag.v36i1.2564>
- Aroyo, L., Taylor, A. S., Diaz, M., Homan, C. M., Parrish, A., Serapio-García, G., Prabhakaran, V., & Wang, D. (2023). DICES Dataset: Diversity in Conversational AI Evaluation for Safety. In *Advances in Neural Information Processing Systems 36: Datasets and Benchmarks Track*. <https://arxiv.org/abs/2306.11247>
- Basile, V., Fell, M., Fornaciari, T., Hovy, D., Paun, S., Plank, B., Poesio, M., & Uma, A. (2021). We Need to Consider Disagreement in Evaluation. In *Proceedings of the 1st Workshop on Benchmarking: Past, Present and Future*, 15–21. Association for Computational Linguistics. <https://doi.org/10.18653/v1/2021.bppf-1.3>
- Fleisig, E., Orlikowski, M., Cimiano, P., & Klein, D. (2025). Balancing Quality and Variation: Spam Filtering Distorts Data Label Distributions. In *Proceedings of the 4th Workshop on Perspectivist Approaches to NLP (NLPerspectives)*, Suzhou, China. <https://aclanthology.org/2025.nlperspectives-1.5/>
- Gordon, M. L., Lam, M. S., Park, J. S., Patel, K., Hancock, J. T., Hashimoto, T., & Bernstein, M. S. (2022). Jury Learning: Integrating Dissenting Voices into Machine Learning Models. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems (CHI '22)*. ACM. <https://doi.org/10.1145/3491102.3502004>
- Homan, C. M., Serapio-García, G., Aroyo, L., Díaz, M., Parrish, A., Prabhakaran, V., Taylor, A. S., & Wang, D. (2024). Intersectionality in AI Safety: Using Multilevel Models to Understand Diverse Perceptions of Safety in Conversational AI. In *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, 131–141. ELRA and ICCL. <https://aclanthology.org/2024.nlperspectives-1.15/>
- Hu, T., & Collier, N. (2024). Quantifying the Persona Effect in LLM Simulations. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 10289–10307. Bangkok, Thailand. <https://aclanthology.org/2024.acl-long.554/>
- Jiang, N.-J., & de Marneffe, M.-C. (2022). Investigating Reasons for Disagreement in Natural Language Inference. *Transactions of the Association for Computational Linguistics*, 10, 1357–1374. https://doi.org/10.1162/tac1_a_00523

- Kirk, H. R., Vidgen, B., Röttger, P., & Hale, S. A. (2024). The Benefits, Risks and Bounds of Personalizing the Alignment of Large Language Models to Individuals. *Nature Machine Intelligence*, 6(4), 383–392. <https://doi.org/10.1038/s42256-024-00820-y>
- Lindahl, A. (2024). Disagreement in Argumentation Annotation. In *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, 56–66. ELRA and ICCL. <https://aclanthology.org/2024.nlperspectives-1.6/>
- Pavlovic, M., & Poesio, M. (2024). The Effectiveness of LLMs as Annotators: A Comparative Overview and Empirical Analysis of Direct Representation. In *Proceedings of the 3rd Workshop on Perspectivist Approaches to NLP (NLPerspectives) @ LREC-COLING 2024*, 100–110. ELRA and ICCL. <https://aclanthology.org/2024.nlperspectives-1.11/>
- Romberg, J. (2022). Is Your Perspective Also My Perspective? Enriching Prediction with Subjectivity. In *Proceedings of the 9th Workshop on Argument Mining*, 115–125. <https://aclanthology.org/2022.argmining-1.11/>
- Sang, Y., & Stanton, J. M. (2022). The Origin and Value of Disagreement Among Data Labelers: A Case Study of Individual Differences in Hate Speech Annotation. In *Proceedings of the 17th International Conference on Information (iConference 2022)*, 425–444. <https://arxiv.org/abs/2112.04030>
- Sorensen, T., Moore, J., Fisher, J., Gordon, M. L., Miresghallah, N., Rytting, C. M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., & Choi, Y. (2024). Position: A Roadmap to Pluralistic Alignment. In *Proceedings of the 41st International Conference on Machine Learning*, PMLR 235, 46280–46302. <https://proceedings.mlr.press/v235/sorensen24a.html>
- Weber-Genzel, L., Peng, S., de Marneffe, M.-C., & Plank, B. (2024). VariErr NLI: Separating Annotation Error from Human Label Variation. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 2256–2269. <https://aclanthology.org/2024.acl-long.123/>

Appendices

The appendices (binary-collapse robustness check, the harm-type grouping scheme, the verbatim prompt texts for both formulations, the nineteen persona definitions, and the reproducibility record) are omitted here for length and are available in the supplementary document and in the project repository.

Reproducibility

All analysis code, computed results, figures and raw model responses are available at:

<https://github.com/rimsha987/disagreemnt-in-nlp->

- Every quantity reported above is generated by the analysis scripts in that repository and listed, with the file that produced it, in `results/SUMMARY_TABLE.md` rather than transcribed by hand.
- The persona prompts, the nineteen persona definitions and the complete raw model responses (180 records, append-only) are included, so the persona arm can be inspected or repeated with a different model.
- The DICES dataset itself is not redistributed. It is released by Google LLC under CC BY 4.0 and is fetched by the setup command given in the repository README.
- The full persona run of 3,675 calls was not executed because the diagnostic stop rule was triggered.